

Yannis FERHAOUI

Software Engineer — 3D — Computer Vision

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Technical Skills

Core & 3D Languages C++, C, Python, GLSL, Qt/QML, Java

Web & Fullstack JavaScript, React.js, PHP/Symfony, HTML/CSS

Specialized Libraries OpenGL, OpenCV, SDL2, VTK/ITK, PyTorch, TensorFlow, NumPy

Engineering & DevOps Git, GitLab CI/CD, Docker, CMake, Linux, GoogleTest

Languages Français (native), Anglais (professional)

Education

Master's in Computer Science – Image, Development and 3D Technologies

2023 – 2025

Claude Bernard University Lyon 1, Villeurbanne

M2 : GPU programming, Virtual Reality, Image and signal processing, Computational geometry.

Tronc commun M1 : Software engineering & project management, Artificial Intelligence, Computational complexity, Design Patterns.

Bachelor's in Computer Science

2020 – 2023

Claude Bernard University Lyon 1, Villeurbanne

Comprehensive CS foundation : Operating systems, IP/TCP networks, SQL databases, Computer architecture, Web development, Advanced algorithms.

Professional Experience

Institut Pascal – Université Clermont Auvergne

Feb. 2025 – Jul. 2025

R&D Intern : Artificial Intelligence & Software Integration

Le Puy-en-Velay

- Architectural integration of LLMs into the open-source medical software 3D Slicer.
- Training and fine-tuning of deep learning models on a multi-GPU compute cluster.
- Development of an extension for user assistance.

AMS Association Mantes Solidarité

May 2023 – Jun. 2023

Fullstack PHP/Symfony Developer

Mantes-La-Ville

- Integration of a payment form and a donation management interface for administrators.

Selected Projects

Mesh Viewer | C++, Qt, OpenGL, CMake

Sep. 2025 – Today

- Design of a high-performance mesh viewer supporting standard formats (.obj, .off).
- Implementation of the graphics pipeline and texture management.
- Setup of a GitLab CI/CD pipeline for automated builds, tests and deployment.

SlicerGPT | Python, Transformers, Qt

Mar. 2025 – Sep. 2025

- 3D Slicer extension integrating a local chatbot.
- Provides context-aware assistance using the scene and the official documentation.

Computational Geometry | C++, Data Structures

Sep. 2024 – Nov. 2024

- Implementation of Lawson's algorithm and mesh manipulation.
- Optimization of data structures for fast processing of point clouds.

Embedded Robotics & Vision | C++, Linux EV3Dev, OpenCV

Feb. 2024 – May 2024

- C++ development on an embedded system for an autonomous robot.
- Real-time image processing for object detection and multi-camera data fusion.
- Management of inter-process communication and motor control.